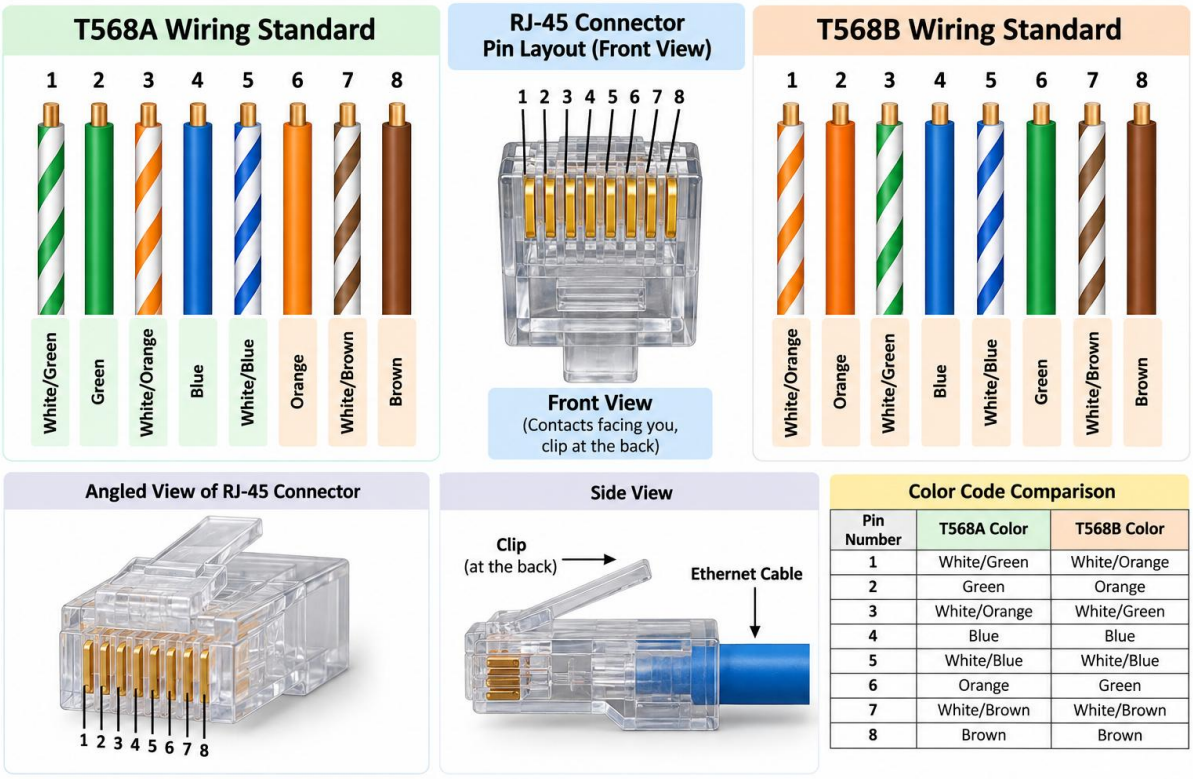
Ethernet Standards and RJ-45 Wiring

1. RJ-45 Connector Pin Layout: T568A and T568B



Pin Number	T568A Color	T568B Color
1	White/Green	White/Orange
2	Green	Orange
3	White/Orange	White/Green
4	Blue	Blue
5	White/Blue	White/Blue
6	Orange	Green
7	White/Brown	White/Brown
8	Brown	Brown

The main difference is that the orange and green wire pairs swap positions between the two standards.

- T568A: White/Green, Green, White/Orange, Blue, White/Blue, Orange, White/Brown, Brown.
- T568B: White/Orange, Orange, White/Green, Blue, White/Blue, Green, White/Brown, Brown.

2. Comparison of Ethernet Standards

Specification	**10BASE-T**	**100BASE-TX**	**1000BASE-T**
Speed	10 Mbps	100 Mbps	1 Gbps (1000 Mbps)

Cable type	Cat3 or better twisted pair	Cat5 or better twisted pair	Cat5e or better twisted pair
Maximum cable length	100 metres	100 metres	100 metres
Connector type	RJ-45 (8P8C)	RJ-45 (8P8C)	RJ-45 (8P8C)
Wire pairs used	2 pairs	2 pairs	All 4 pairs
Typical use	Older Ethernet networks	Fast Ethernet LANs	Modern home and college LANs

3. Connecting Two Laptops Directly Using Ethernet

Traditionally, a crossover cable is used to connect two laptops directly because it swaps the transmit and receive pairs.

- In a straight-through cable, both ends use the same wiring standard, such as T568B at both ends.
- In a crossover cable, one end uses T568A and the other uses T568B, crossing the relevant transmit and receive pairs.
- However, most modern laptops and Ethernet adapters support Auto MDI-X, which automatically adjusts the transmit and receive pairs. Therefore, a straight-through cable will usually work too.

Conclusion: A crossover cable is the traditional answer for a direct laptop-to-laptop connection, but with modern devices, either cable type will generally work if Auto MDI-X is supported.

4. Three Devices That May Use Ethernet at Home or College

Device	**Likely Ethernet Standard**	**Reason**
Desktop computer or gaming PC	1000BASE-T	Supports high-speed file transfers, gaming and Internet access.
Wi-Fi router	1000BASE-T	Gigabit LAN ports connect wired devices at high speeds.
Network printer	100BASE-TX or 1000BASE-T	Provides a wired connection for sharing printing services across a LAN.

Note: The actual Ethernet standard depends on the device's hardware specifications. Check the manufacturer's documentation to confirm the supported speed.